

JDBC Connection to MySQL (music_db)

Q1: Install MySQL Community Server and Connector/J

Step 1: Install MySQL Community Server

1. Download the MySQL Installer (Windows) or the server package for your OS from the MySQL Community Downloads page (dev.mysql.com/downloads).
2. Run the installer, choose "Server only" or "Developer Default", and set the root password (use root123 to match the questions).
3. Finish the setup and make sure the MySQL service is running on port 3306.
4. Create the database:

```
CREATE DATABASE music_db;
```

Step 2: Download the JDBC driver (Connector/J)

1. On the same downloads page, choose "Connector/J" and select "Platform Independent".
2. Download the ZIP and extract it.
3. The file you need is `mysql-connector-j-x.x.x.jar`.

Step 3: Write down the jar file path

Write the path where you saved the jar. Example on Windows:

```
C:\jdbc\mysql-connector-j-x.x.x.jar
```

Example on Linux/Mac:

```
/home/username/jdbc/mysql-connector-j-x.x.x.jar
```

Replace x.x.x with the version you downloaded, and write your own real path in the assignment.

Q2: TestDBConnection.java

```
// TestDBConnection.java
// Connects to the local MySQL database 'music_db' using JDBC DriverManager.

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class TestDBConnection {
    public static void main(String[] args) {
        String url      = "jdbc:mysql://localhost:3306/music_db";
        String username = "root";
        String password = "root123";

        try {
            // Load the MySQL JDBC driver (optional in newer versions, but safe to include)
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Establish the connection using DriverManager
            Connection con = DriverManager.getConnection(url, username, password);

            if (con != null) {
                System.out.println("Connection Successful");
                con.close();
            }
        } catch (ClassNotFoundException e) {
            System.out.println("JDBC Driver not found: " + e.getMessage());
        } catch (SQLException e) {
            System.out.println("SQL Error: " + e.getMessage());
        }
    }
}
```

How to compile and run (use your own jar path). Windows:

```
javac TestDBConnection.java
java -cp ".;C:\jdbc\mysql-connector-j-x.x.x.jar" TestDBConnection
```

Linux/Mac:

```
javac TestDBConnection.java
java -cp ".:~/home/username/jdbc/mysql-connector-j-x.x.x.jar" TestDBConnection
```

Expected output:

Connection Successful

Q3: Wrong password and the error message

Change the password in the code to a wrong one, and add the explanation as a comment:

```
String password = "wrong123"; // intentionally wrong password

/*
 * OBSERVED ERROR:
 * SQL Error: Access denied for user 'root'@'localhost' (using password: YES)
 *
 * EXPLANATION:
 * The MySQL server rejected the login because the password sent by the program
 * does not match the password stored for the user 'root'.
 * "using password: YES" means a password was supplied, but it was incorrect.
 * DriverManager.getConnection() threw a SQLException (SQLState 28000, error code 1045),
 * which was caught in the catch block and printed. The connection was not created.
 */
```

Output when you run it:

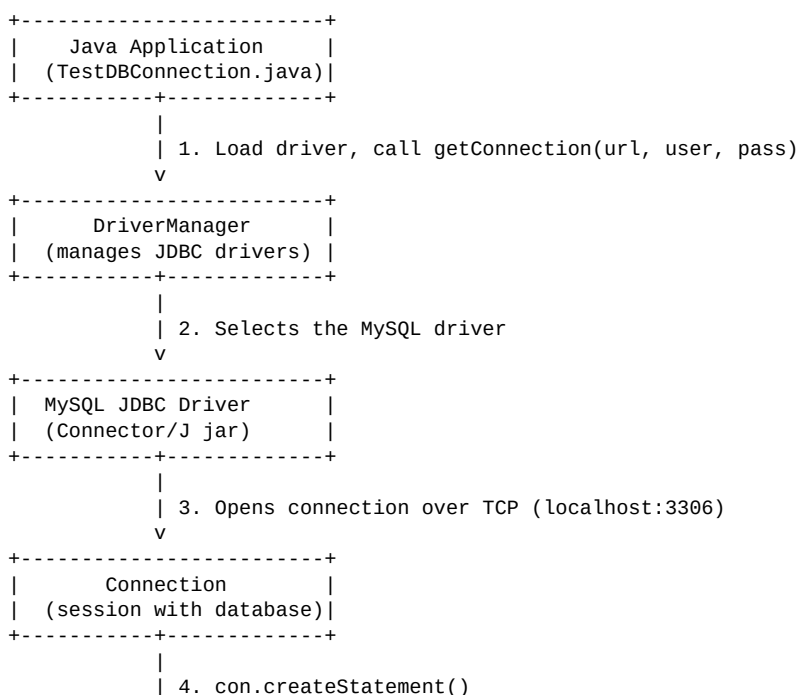
SQL Error: Access denied for user 'root'@'localhost' (using password: YES)

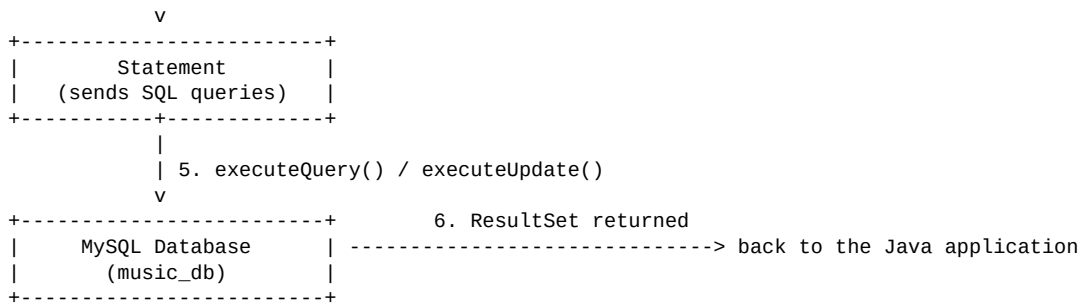
Your exact message may vary slightly by MySQL and driver version, so copy what appears on your own screen. If you also want to see the error code, add this line inside the catch block:

```
System.out.println("SQLState: " + e.getSQLState() + ", Error Code: " + e.getErrorCode());
```

Q4: JDBC connection diagram

Copy this diagram onto paper or into a tool like draw.io, then take a photo or screenshot to submit.





Labels to include in your drawing:

- Java Application
- DriverManager
- MySQL JDBC Driver (Connector/J)
- Connection
- Statement
- MySQL Database (music_db)
- ResultSet (returned to the application)
- Arrows numbered 1 to 6 as shown above